

THESIS SAMPLE • SUBJECT COURSE: INFORMATION
TECHNOLOGY

IMPLEMENTATION OF MACHINE LEARNING-BASED INTRUSION DETECTION SYSTEM FOR SMALL AND MEDIUM ENTERPRISES IN THE PHILIPPINES

REDACTED

MASTER OF SCIENCE IN INFORMATION TECHNOLOGY | MASTERAL
LEVEL

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CHAPTER 1

THE PROBLEM AND ITS BACKGROUND

Introduction

The rapid advancement of information and communication technology has fundamentally transformed the operational landscape of businesses across the globe. In the Philippines, small and medium enterprises (SMEs) constitute approximately 99.5% of all registered businesses and contribute nearly 35% of the country's gross domestic product, according to the Department of Trade and Industry (DTI, 2023). As these enterprises increasingly adopt digital platforms, cloud-based systems, and interconnected networks to streamline their operations, they simultaneously expose themselves to a growing spectrum of cybersecurity threats that have become more sophisticated, frequent, and damaging in recent years.

Cyber intrusions, unauthorized access attempts, network breaches, malware infiltrations, and denial-of-service attacks, have escalated at an alarming rate both globally and within the Philippine context. The Cybercrime Investigation and Coordinating Center (CICC) reported a 43% increase in cybercrime incidents in the Philippines between 2021 and 2023, with SMEs accounting for a disproportionately large share of victims due to their limited cybersecurity infrastructure and relatively low investment in digital defense mechanisms (CICC, 2023). Unlike large corporations that maintain dedicated IT security teams and enterprise-

grade protection systems, most Philippine SMEs operate with minimal technical personnel and rely on basic, often outdated, security measures that are ill-equipped to detect and respond to modern cyber threats in real time.

Traditional intrusion detection systems (IDS) have long served as a frontline defense mechanism in network security architectures. These systems monitor network traffic and system activities for signs of malicious behavior, alerting administrators to potential threats before significant damage occurs. However, conventional rule-based and signature-based IDS solutions carry inherent limitations that render them increasingly inadequate in the face of evolving attack methodologies. Specifically, they are incapable of identifying novel or zero-day threats for which no predefined signatures exist, require constant manual updating of rule sets, and generate high volumes of false positives that burden already stretched IT resources, a particularly acute problem for SMEs operating with lean technical teams.

Machine learning (ML), a subset of artificial intelligence, has emerged as a transformative approach to intrusion detection that addresses many of the shortcomings of traditional systems. By training algorithms on large datasets of network behavior, ML-based IDS solutions can learn to distinguish between normal and anomalous traffic patterns without relying on predefined rules, enabling the detection of previously unseen attack vectors with significantly greater accuracy and adaptability. Studies conducted in international contexts have demonstrated the superiority of ML-based intrusion detection over conventional methods across multiple performance metrics, including detection rate, false positive rate, and response latency (Buczak and Guven, 2016; Liao et al., 2013; Vinayakumar et al., 2019).

Despite the demonstrated effectiveness of machine learning approaches in cybersecurity research, their adoption among Philippine SMEs remains negligible. This gap exists due to a combination of factors including the high cost of enterprise ML security platforms, the technical complexity of implementation and maintenance, a shortage of cybersecurity professionals with ML expertise in the local market, and a general lack of awareness among SME owners and operators regarding the capabilities and accessibility of ML-based solutions. As a consequence, the majority of Philippine SMEs remain dangerously exposed to cyber intrusions that could be detected, and in many cases prevented, by appropriately designed and deployed machine learning systems.

This study seeks to address this critical gap by designing, implementing, and evaluating a machine learning-based intrusion detection system specifically tailored to the technical constraints, operational realities, and resource limitations of small and medium enterprises in the Philippines. By developing a solution that is both effective and accessible within the Philippine SME context, this research aims to contribute meaningfully to the nation's cybersecurity resilience and to the protection of enterprises that form the backbone of the Philippine economy.

Background of the Study

The cybersecurity landscape in the Philippines has undergone significant deterioration over the past decade, driven by the country's accelerating digital transformation and the corresponding expansion of its attack surface. The Philippines ranked among the top ten most targeted countries for cyberattacks in Southeast Asia in 2022, with financial institutions, government agencies, healthcare providers, and commercial enterprises all recorded as frequent targets (Kaspersky Security Bulletin,

2022). The economic impact of these incidents extends far beyond immediate financial losses, encompassing reputational damage, operational disruption, regulatory penalties, and the erosion of customer trust.

For Philippine SMEs specifically, the consequences of a successful cyber intrusion can be devastating and potentially irreversible. A 2022 survey conducted by the Philippine Chamber of Commerce and Industry (PCCI) found that 60% of SMEs that experienced a significant cybersecurity breach reported financial losses exceeding 500,000 pesos, while 23% were forced to temporarily suspend operations. More critically, 18% of breached SMEs reported permanent closure within twelve months of the incident, underscoring the existential threat that inadequate cybersecurity postures represent for smaller enterprises (PCCI, 2022).

The nature of cyber threats targeting Philippine SMEs has evolved considerably in recent years. Early incidents were dominated by relatively unsophisticated attacks such as phishing emails and basic malware infections. Contemporary threats, however, increasingly involve advanced persistent threats (APTs), ransomware campaigns, supply chain attacks, and AI-assisted intrusion techniques that adapt dynamically to evade detection. The 2023 Philippine National Cybersecurity Plan acknowledged this evolution, noting that threat actors are employing increasingly automated and intelligent attack methodologies that outpace the detection capabilities of conventional security tools (Department of Information and Communications Technology, 2023).

Machine learning has been at the forefront of academic and industry responses to this evolving threat landscape. Pioneering research by Tavallaee et al. (2009) demonstrated that ML algorithms could achieve detection rates exceeding 99% on benchmark intrusion datasets, a finding that has since been replicated and extended by numerous subsequent studies. More recent work has focused on deep learning architectures,

including convolutional neural networks (CNNs), recurrent neural networks (RNNs), and long short-term memory (LSTM) networks, which have shown particular promise in detecting complex, multi-stage intrusion patterns that unfold over extended time periods (Kim et al., 2016; Staudemeyer and Omlin, 2014).

However, a persistent gap exists between the performance demonstrated in controlled research environments and the practical deployment of ML-based IDS solutions in real-world SME settings. Most existing implementations have been designed for enterprise environments with abundant computational resources, large and continuously updated training datasets, and dedicated security operations teams capable of managing and interpreting ML model outputs. The adaptation of these approaches to resource-constrained SME environments, particularly within the specific technological and economic context of the Philippines, remains an underexplored area of research that this study directly addresses.

Statement of the Problem

This study seeks to design and implement a machine learning-based intrusion detection system appropriate for deployment within Philippine small and medium enterprises. Specifically, it aims to address the following research problems:

1. What is the current state of cybersecurity preparedness among small and medium enterprises in the Philippines, particularly with respect to intrusion detection capabilities?
2. What machine learning algorithm or combination of algorithms yields the highest detection accuracy, lowest false positive rate, and optimal computational efficiency for intrusion detection within the resource constraints typical of Philippine SMEs?

3. How does the performance of the proposed ML-based intrusion detection system compare to that of conventional signature-based and rule-based intrusion detection systems when evaluated on Philippine SME network traffic data?

4. To what extent does the implementation of the proposed ML-based IDS improve the overall cybersecurity posture of participating SMEs as measured by detection rate, response time, and reduction in successful intrusion incidents?

5. What implementation framework, inclusive of technical specifications, deployment procedures, and personnel training requirements, is most appropriate for the adoption of ML-based intrusion detection within the Philippine SME context?

Significance of the Study

The findings of this study are expected to yield substantial benefits for multiple stakeholder groups within the Philippine technology, business, and policy ecosystem.

Small and Medium Enterprises. SMEs stand to benefit most directly from this research through access to an evidence-based, practically validated framework for implementing machine learning-based intrusion detection that is calibrated to their specific resource constraints and operational realities. By following the implementation model developed in this study, SMEs can significantly enhance their ability to detect and respond to cyber intrusions before they result in material harm.

Information Technology Practitioners and Cybersecurity Professionals. This study provides IT professionals serving the SME market with a technically rigorous, context-specific guide to ML-based IDS deployment. The comparative performance analysis of multiple machine learning algorithms offers practitioners a data-driven basis for algorithm selection in similar deployment scenarios.

Academic Researchers. This research contributes to the body of knowledge on applied machine learning in cybersecurity within the Southeast Asian context, an area that remains underrepresented in international literature relative to its practical significance. The dataset, methodology, and findings generated by this study provide a foundation for subsequent research into advanced intrusion detection approaches for developing market environments.

Government and Policy Makers. Findings from this study can inform the development of national cybersecurity standards and incentive programs specifically designed to elevate the security posture of Philippine SMEs, contributing to the objectives outlined in the Philippine National Cybersecurity Plan 2023-2028.

Educational Institutions. The study provides a practical case study and reference material for information technology and cybersecurity programs, bridging the gap between theoretical machine learning concepts and their applied implementation in real-world security contexts.

Scope and Delimitation

This study focuses on the design, implementation, and evaluation of a machine learning-based intrusion detection system for deployment within small and medium enterprises operating in the Philippines. The research is delimited by the following parameters.

The study covers network-based intrusion detection, the monitoring and analysis of network traffic data, and does not extend to host-based intrusion detection systems that monitor individual device behavior, nor to endpoint detection and response (EDR) platforms. The research is further limited to the detection of intrusion attempts and does not encompass automated intrusion prevention or active threat response mechanisms.

Machine learning algorithms evaluated in this study are limited to supervised and semi-supervised learning approaches applicable to labeled network traffic datasets, specifically Random Forest, Support Vector Machine (SVM), Gradient Boosting, and Long Short-Term Memory (LSTM) neural network architectures. Unsupervised learning approaches and reinforcement learning models are acknowledged as areas for future research but are outside the scope of this study.

The empirical component of this research involves SME participants drawn from Metro Davao, representing a range of industries including retail, services, manufacturing, and professional services. Findings are therefore most directly applicable to SMEs operating in comparable urban Philippine settings, though the general implementation framework is designed to be broadly applicable across the Philippine SME sector.

The study is conducted within the period of 2024 to 2025 and reflects the cybersecurity threat landscape and available machine learning tools and frameworks as of that period.

Definition of Terms

The following terms are defined as they are used throughout this study.

Anomaly Detection. A method of intrusion detection that identifies network behavior deviating significantly from an established baseline of normal activity, enabling the identification of novel or previously unseen attack types.

Cyber Intrusion. Any unauthorized attempt to access, disrupt, or exfiltrate data from a computer system or network, including but not limited to hacking, malware deployment, phishing, and denial-of-service attacks.

False Positive Rate. The proportion of legitimate network traffic incorrectly classified as malicious by an intrusion detection system, expressed as a percentage of total benign traffic samples.

Gradient Boosting. An ensemble machine learning technique that builds a predictive model by sequentially combining multiple weak learners, typically decision trees, with each successive learner trained to correct the errors of its predecessors.

Intrusion Detection System (IDS). A software application or hardware device that monitors network or system activities for malicious behavior or policy violations and generates alerts upon detection of suspicious activity.

Long Short-Term Memory (LSTM). A type of recurrent neural network architecture capable of learning long-term dependencies in sequential data, particularly suited to the detection of multi-stage intrusion patterns that unfold across extended time sequences.

Machine Learning (ML). A branch of artificial intelligence in which algorithms are trained on datasets to identify patterns and make predictions or decisions without being explicitly programmed with rules for each specific scenario.

Network Traffic. The data transmitted between devices over a computer network, including packet headers, payload content, connection metadata, and timing information used as input features for ML-based intrusion detection models.

Random Forest. An ensemble learning method that constructs multiple decision trees during training and outputs the class most frequently predicted across all trees, offering high accuracy and resistance to overfitting.

Small and Medium Enterprise (SME). As defined by the Philippine Department of Trade and Industry, a business entity with total assets between 3,000,001 pesos and 100,000,000 pesos and employing between ten and 199 employees.

Support Vector Machine (SVM). A supervised machine learning algorithm that identifies the optimal hyperplane separating data points of different classes in high-dimensional space, commonly applied to binary classification problems in intrusion detection.

Zero-Day Threat. A cyber attack that exploits a previously unknown vulnerability for which no patch or signature-based defense currently exists, rendering conventional signature-based intrusion detection systems incapable of identification.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

Introduction

This chapter presents a comprehensive review of literature and studies relevant to the present research. The review is organized into four major sections: foreign literature, local literature, foreign studies, and local studies. Each section examines existing knowledge pertaining to cybersecurity threats, intrusion detection systems, machine learning applications in network security, and the cybersecurity challenges specific to small and medium enterprises. The chapter concludes with a synthesis that identifies the gaps in existing knowledge that this study seeks to address.

Foreign Literature

Cybersecurity Threats and the Evolving Attack Landscape

The global cybersecurity landscape has undergone profound transformation over the past two decades, driven by the exponential growth of internet-connected devices, the migration of critical business operations to digital platforms, and the increasing sophistication of threat actors operating across national boundaries. Stallings and Brown (2018) describe the contemporary threat environment as characterized by a fundamental asymmetry between attackers and defenders: while defenders

must successfully protect all potential attack surfaces simultaneously, attackers need only identify and exploit a single vulnerability to achieve their objectives. This asymmetry is particularly acute for organizations with limited security resources, such as SMEs.

The IBM Security X-Force Threat Intelligence Index (2023) identified phishing, exploitation of public-facing applications, and supply chain compromise as the three most prevalent initial attack vectors globally, collectively accounting for 67% of all recorded incidents. More significantly, the report noted a 41% year-on-year increase in the deployment of backdoors, persistent access mechanisms installed by attackers to maintain covert access to compromised systems, indicating a strategic shift among threat actors toward long-duration, low-visibility intrusions designed to extract maximum value before detection. Such advanced persistent threats pose particular challenges for conventional detection approaches reliant on known attack signatures.

Verizon's Data Breach Investigations Report (2023) corroborates this trend, finding that 74% of all breaches involve a human element, including social engineering, errors, or misuse, while 83% of breaches are financially motivated. The report further establishes that SMEs experience breach patterns fundamentally different from those affecting large enterprises, with SMEs disproportionately targeted by ransomware, credential theft, and basic web application attacks, categories for which rapid, automated detection is both most impactful and most feasible through machine learning approaches.

Traditional Intrusion Detection Systems

The concept of automated intrusion detection was formally articulated by Anderson (1980) in a landmark technical report for the United States Air Force, which proposed the monitoring of system audit trails for patterns indicative of unauthorized use. This foundational work established the conceptual framework for IDS development that persists,

in evolved form, to the present day. Denning (1987) subsequently formalized the statistical anomaly detection model, proposing that deviations from established behavioral baselines could serve as reliable indicators of intrusion, regardless of the specific attack technique employed.

Contemporary IDS architectures are broadly categorized into two primary types based on detection methodology. Signature-based systems, also termed misuse detection systems, identify intrusions by comparing observed network traffic or system behavior against a database of known attack patterns or signatures. These systems offer high accuracy and low false positive rates for known attack types but are fundamentally incapable of detecting novel threats for which no signature has been developed (Liao et al., 2013). Given the rate at which new malware variants and attack techniques emerge, with AV-TEST Institute registering over 450,000 new malicious programs daily as of 2023, the limitations of signature-based detection have become increasingly consequential.

Anomaly-based systems, by contrast, establish statistical models of normal network behavior and flag deviations from these baselines as potential intrusions. While theoretically capable of detecting novel attack types, early anomaly-based systems suffered from unacceptably high false positive rates that generated alert fatigue among security personnel and undermined confidence in automated detection (Axelsson, 2000). The challenge of balancing detection sensitivity against false positive generation, formally described as the detection-false alarm tradeoff, remains one of the central technical challenges in IDS design, and one that machine learning approaches have demonstrated particular promise in addressing.

Machine Learning Foundations in Cybersecurity

The application of machine learning to network intrusion detection has been an active area of research since the late 1990s, gaining substantial momentum with the publication of the KDD Cup 1999 dataset, a benchmark collection of labeled network traffic data that enabled standardized comparative evaluation of ML-based detection approaches (Tavallae et al., 2009). While subsequent research has identified significant limitations in the KDD Cup dataset, including class imbalances and feature redundancies, its publication catalyzed a generation of research demonstrating the fundamental viability of machine learning for intrusion detection.

Buczak and Guven (2016) conducted a comprehensive survey of machine learning and data mining methods applied to cybersecurity, reviewing 106 papers published between 2000 and 2015. Their analysis identified decision tree-based methods, artificial neural networks, and support vector machines as the most frequently employed approaches, with ensemble methods, particularly Random Forest, demonstrating consistently superior performance across multiple evaluation metrics. The survey concluded that machine learning approaches consistently outperformed rule-based systems on detection rate while exhibiting considerable variability in false positive performance, identifying false positive minimization as the most significant remaining challenge for practical deployment.

Chandola, Banerjee, and Kumar (2009) provide a theoretical framework for anomaly detection across multiple domains, categorizing detection techniques according to their underlying assumptions about normal behavior and the types of anomalies they are designed to identify. Their taxonomy distinguishes between point anomalies, individual data instances that deviate from the norm, contextual anomalies, instances that are anomalous only in a specific context, and collective anomalies,

collections of instances that are anomalous as a group but individually normal. This is a distinction with direct practical relevance to network intrusion detection, where many sophisticated attack patterns manifest as collective rather than point anomalies.

Random Forest in Intrusion Detection

Random Forest, introduced by Breiman (2001), is an ensemble learning method that constructs a multitude of decision trees during training and outputs the mode of their individual classifications. Its application to intrusion detection has been extensively validated across multiple studies and benchmark datasets. Farnaaz and Jabbar (2016) evaluated Random Forest against multiple competing algorithms on the NSL-KDD dataset, finding that Random Forest achieved a detection accuracy of 99.67% while maintaining a false positive rate of 0.4%, performance metrics that substantially exceeded those of single decision tree, Naive Bayes, and k-Nearest Neighbor classifiers evaluated under identical conditions.

The robustness of Random Forest to noise and its ability to handle high-dimensional feature spaces without explicit feature selection make it particularly well-suited to network traffic classification, where raw packet data may contain hundreds of features of varying relevance and quality. Additionally, Random Forest's inherent feature importance ranking capability provides security analysts with interpretable insights into which network traffic characteristics most strongly predict malicious activity, a practically valuable property that pure neural network approaches often lack (Liaw and Wiener, 2002).

Support Vector Machines in Network Security

Support Vector Machines (SVMs), developed by Cortes and Vapnik (1995), identify the optimal hyperplane separating data points of different classes in high-dimensional feature space. Their application to intrusion detection has demonstrated strong performance, particularly in

scenarios involving binary classification between normal and anomalous traffic. Shon and Moon (2007) applied SVM to network intrusion detection in combination with genetic algorithm-based feature selection, achieving a detection rate of 99.35% with a false positive rate of 0.02% on the KDD Cup 1999 dataset, one of the lowest false positive rates reported for that benchmark.

The kernel trick, which enables SVMs to operate efficiently in very high-dimensional feature spaces by computing dot products in the original space rather than the transformed space, allows these algorithms to capture complex, nonlinear decision boundaries that characterize the distinction between legitimate and malicious network traffic. However, SVMs exhibit known computational scaling challenges with large training datasets, as training time complexity grows quadratically with the number of samples, a consideration relevant to deployment in environments where continuous retraining on updated traffic data is operationally necessary (Scholkopf and Smola, 2002).

Deep Learning and LSTM Networks

The application of deep learning architectures to intrusion detection represents the current frontier of research in the field. Recurrent neural networks (RNNs), and particularly Long Short-Term Memory (LSTM) networks introduced by Hochreiter and Schmidhuber (1997), offer unique capabilities for analyzing sequential network traffic data by maintaining internal state across time steps, enabling the detection of attack patterns that unfold across multiple packets or connection events rather than manifesting in any single isolated observation.

Kim, Kim, and Kim (2016) applied LSTM networks to the detection of network intrusions, demonstrating that the architecture's ability to capture temporal dependencies in traffic sequences enabled superior detection of slow, distributed attack patterns, including slow-rate denial-of-service attacks and staged APT behaviors, that evade detection

by algorithms that analyze individual connections in isolation. Their experimental results showed LSTM achieving a detection accuracy of 93.82% on a dataset specifically constructed to include temporally distributed attacks, compared to 69.41% for a conventional feedforward neural network evaluated under identical conditions.

More recent work by Garg et al. (2021) proposed hybrid deep learning architectures combining convolutional neural networks (CNNs) for spatial feature extraction from raw packet data with LSTM layers for temporal pattern recognition, achieving state-of-the-art detection rates on the CICIDS2017 and UNSW-NB15 benchmark datasets. While such hybrid approaches represent the performance frontier in controlled research settings, their computational requirements and architectural complexity present implementation challenges that must be carefully evaluated against the resource constraints characteristic of SME deployment environments.

Local Literature

Philippine Cybersecurity Policy and Regulatory Framework

The legal and regulatory framework governing cybersecurity in the Philippines has developed substantially over the past decade, providing an increasingly formalized context within which SME cybersecurity obligations and resources must be understood. Republic Act 10175, the Cybercrime Prevention Act of 2012, established the foundational legal basis for cybercrime prosecution in the Philippines, defining offenses including illegal access, data interference, system interference, and cyber-squatting, while creating institutional mechanisms for cybercrime investigation and prosecution through the Department of Justice and the National Bureau of Investigation.

The enactment of Republic Act 10173, the Data Privacy Act of 2012, introduced data protection obligations directly relevant to SME cybersecurity practices. The Act mandates that personal information controllers, a category that encompasses virtually all businesses collecting customer or employee data, implement reasonable and appropriate organizational, physical, and technical security measures to protect personal information against natural dangers and human dangers including accidental or unlawful destruction, alteration, and unauthorized disclosure or access. The National Privacy Commission (NPC) has issued implementing rules and data breach notification requirements that establish minimum security standards applicable to Philippine SMEs (NPC, 2021).

The Department of Information and Communications Technology (DICT) has articulated national cybersecurity strategy through the Philippine National Cybersecurity Plan 2023-2028, which identifies the protection of critical information infrastructure, the development of cybersecurity awareness, and the enhancement of the national cyber incident response capability as primary strategic priorities. Notably, the plan explicitly acknowledges SMEs as a high-risk population requiring targeted cybersecurity support, citing their limited resources and technical capacity as factors rendering them disproportionately vulnerable to cyber threats (DICT, 2023).

Digital Transformation and Technology Adoption Among Philippine SMEs

The Philippine SME sector's engagement with digital technology has accelerated substantially in the years following the COVID-19 pandemic, driven by the necessity of remote operations, e-commerce adoption, and digital payment integration during periods of physical distancing restrictions. The DTI's MSME Development Plan 2022-2028

documents this acceleration, reporting a 78% increase in SME digital platform adoption between 2020 and 2022, with e-commerce, digital payments, and cloud-based productivity tools representing the most rapidly adopted categories (DTI, 2022).

However, this accelerated digitalization has not been accompanied by commensurate investment in cybersecurity infrastructure. Asuncion and Villanueva (2021) surveyed 215 SMEs in Metro Manila and found that only 23% had implemented any form of dedicated cybersecurity solution beyond basic antivirus software, while 67% reported having no formal cybersecurity policy or incident response procedure. Critically, 81% of respondents indicated that cybersecurity investment was not a priority in their annual budget planning, with the majority citing cost, lack of perceived necessity, and insufficient technical expertise as the primary barriers to adoption.

This gap between digital adoption and cybersecurity investment creates precisely the conditions under which machine learning-based intrusion detection offers the greatest potential impact: a growing digital attack surface, limited human security resources, and the need for automated, low-maintenance detection solutions that do not require constant expert attention to remain effective.

Foreign Studies

Comparative Studies of ML Algorithms for Intrusion Detection

Sultana et al. (2019) conducted a systematic comparative evaluation of seven machine learning algorithms, including Random Forest, SVM, Decision Tree, K-Nearest Neighbor, Naive Bayes, Logistic Regression, and Artificial Neural Network, applied to the UNSW-NB15 intrusion detection dataset. Their findings established Random Forest as the highest-performing algorithm across the composite evaluation metric incorporating accuracy, false positive rate, and computational efficiency,

achieving 97.04% accuracy with a false positive rate of 1.8%. The study further identified feature selection as a critical preprocessing step, with optimal feature subsets varying significantly across algorithm types and attack categories.

Vinayakumar et al. (2019) presented a comprehensive deep learning framework for network intrusion detection evaluated across multiple benchmark datasets including KDD Cup 1999, NSL-KDD, UNSW-NB15, and CICIDS2017. Their comparative analysis demonstrated that deep learning architectures, particularly recurrent and convolutional variants, substantially outperformed traditional ML algorithms on datasets containing temporally distributed and polymorphic attack patterns, while performing comparably on datasets dominated by well-defined, discrete attack signatures. This finding has direct implications for algorithm selection in deployment environments where the anticipated threat mix includes both conventional and advanced persistent threats.

Yin et al. (2017) specifically investigated the application of LSTM networks to intrusion detection on the KDD Cup 1999 dataset, achieving a detection accuracy of 93.82% and demonstrating that the recurrent architecture's temporal memory enabled superior detection of U2R (user-to-root) and R2L (remote-to-local) attack categories, attack types that involve multi-stage sequences of actions and are among the most difficult for non-sequential algorithms to detect. Their work established a foundational benchmark for LSTM-based intrusion detection against which subsequent studies, including the present research, can be evaluated.

SME-Focused Cybersecurity Research

Ngoqo and Flowerday (2015) examined information security behaviors among SME employees in developing economies, finding that security awareness, perceived severity of threats, and organizational security culture were the most significant predictors of security-compliant behavior. Their research highlights the importance of considering human and organizational factors, not only technical solutions, in SME cybersecurity interventions, a consideration that informs the training and change management components of the implementation framework proposed in this study.

Bada and Sasse (2014) analyzed cybersecurity awareness program effectiveness across multiple organizational contexts and found that awareness programs achieve greatest behavior change when they present concrete, relevant threat scenarios rather than abstract risk frameworks. This finding supports the integration of locally-relevant Philippine threat examples in the personnel training component of the proposed implementation framework, ensuring that SME staff can recognize and appropriately respond to the specific threat patterns most commonly directed at Philippine businesses.

Micro Focus (2022) conducted a global survey of SME cybersecurity practices across 15 countries, finding that SMEs with annual revenues below \$10 million spent an average of 7.5% of their IT budget on cybersecurity, approximately one-third of the proportion invested by large enterprises. The survey further identified automated threat detection as the single cybersecurity capability most frequently cited by SME owners as their highest priority investment need, directly validating the focus of the present research on machine learning-based automated intrusion detection.

Local Studies

Cybersecurity Incident Research in the Philippine Context

Yuchengco (2022) analyzed 847 cybersecurity incidents reported to the CICC between 2019 and 2021, identifying retail and wholesale trade SMEs as the most frequently victimized sector, accounting for 31% of all recorded incidents. The study found that the median time from initial intrusion to detection among SME victims was 197 days, far exceeding the global median of 24 days reported by Mandiant (2022) for organizations with dedicated security operations capabilities. This detection lag finding provides compelling quantitative evidence for the critical importance of automated intrusion detection among Philippine SMEs and establishes a baseline against which the detection timeliness improvements enabled by the proposed system can be measured.

Reyes and Cacanindin (2020) investigated the specific cybersecurity vulnerabilities prevalent in Philippine SME network infrastructures, finding that misconfigured network devices, use of default credentials, unpatched operating systems, and absence of network segmentation were the four most commonly observed vulnerability categories. These findings directly inform the feature engineering component of the ML model developed in this study, as network traffic patterns associated with exploitation of these specific vulnerability categories are prioritized in the model's training dataset construction.

Bautista and Hernandez (2021) evaluated the feasibility of cloud-based security solutions for Philippine SMEs, finding that cloud-delivered security services offered a cost-effective pathway to enterprise-grade protection for resource-constrained businesses. Their cost-benefit analysis, conducted across a sample of 48 Cebu-based SMEs, found that cloud-based security implementations achieved positive return on investment within an average of 14 months, primarily through avoidance of incident

response and recovery costs. This finding supports the cloud-compatible deployment architecture proposed in this research, which is designed to enable SMEs to leverage cloud infrastructure to overcome local computational resource limitations.

Taguinod et al. (2023) conducted the most recent and directly relevant local study, implementing a Random Forest-based intrusion detection system in three Metro Manila SMEs across a six-month operational period. Their results demonstrated a 94.3% detection rate for known attack types and an 81.7% detection rate for novel attack variants, with a false positive rate of 3.2%. Critically, the study documented substantial practical barriers encountered during implementation, including resistance from non-technical staff, difficulty integrating the IDS with legacy network equipment, and the absence of local technical support resources for troubleshooting. These documented implementation challenges directly inform the practical framework proposed in the present study, which incorporates specific provisions for staff change management, legacy equipment compatibility, and local technical support capacity building.

Synthesis of the Review

The foregoing review of related literature and studies reveals a convergence of evidence supporting both the critical need for and the technical feasibility of machine learning-based intrusion detection systems within the Philippine SME context. Several overarching themes emerge from this synthesis that directly inform the design and methodology of the present research.

First, the evidence consistently establishes that Philippine SMEs face a severe and worsening cybersecurity threat environment characterized by increasing attack frequency, growing attack sophistication, and detection timelines averaging 197 days, which are

catastrophically long relative to the damage potential of modern cyber intrusions. This threat environment, combined with the demonstrated inadequacy of conventional signature-based detection against novel attack types, creates an urgent and well-documented need for the type of adaptive, automated detection capability that machine learning systems provide.

Second, comparative studies of ML algorithms for intrusion detection consistently identify ensemble methods, particularly Random Forest, as offering the optimal balance of detection accuracy, false positive performance, computational efficiency, and interpretability for practical deployment. LSTM networks demonstrate superior performance specifically for temporally distributed attack patterns, suggesting that a hybrid or ensemble approach incorporating both Random Forest and LSTM components may offer performance advantages over either algorithm deployed independently. This finding directly informs the multi-algorithm evaluation design of the present study.

Third, existing local studies, while confirming the technical viability of ML-based IDS solutions in the Philippine context, document significant implementation barriers related to organizational, technical, and resource factors that pure technical studies have largely overlooked. The practical implementation framework proposed in this research directly addresses these documented barriers, positioning the present study to make a contribution that extends beyond technical performance evaluation to encompass the practical knowledge needed for actual SME deployment.

Finally, a notable gap exists in the existing literature with respect to the development and validation of ML-based IDS solutions specifically designed for and evaluated within the Philippine SME context, using locally representative network traffic data reflecting the specific attack

patterns and normal traffic characteristics of Philippine business networks. The present study addresses this gap directly, contributing both a technically evaluated system and a contextually grounded implementation framework to the growing body of Philippine cybersecurity research.

CHAPTER 3

RESEARCH METHODOLOGY

Introduction

This chapter presents the research design, locale of the study, participants, instrumentation, data collection procedures, system development methodology, and statistical treatment employed in this investigation. The methodological framework is designed to ensure the scientific rigor, replicability, and practical applicability of the research findings within the Philippine SME context. The chapter further addresses the ethical considerations observed throughout the conduct of the study to protect the rights, privacy, and interests of all participating organizations and individuals.

Research Design

This study employs a mixed-methods research design combining quantitative experimental evaluation with qualitative descriptive analysis. The quantitative component encompasses the empirical development, training, and performance evaluation of machine learning models for network intrusion detection, utilizing controlled experimental procedures to generate objective, measurable performance data across multiple

evaluation metrics. The qualitative component involves the descriptive analysis of implementation experiences, organizational barriers, and practitioner perceptions gathered from SME participants during the system deployment phase.

The primary research approach follows the Design Science Research (DSR) paradigm as formalized by Hevner et al. (2004), which provides a rigorous framework for the development and evaluation of information technology artifacts, in this case, the ML-based intrusion detection system, within their intended organizational context. DSR is particularly appropriate for applied computing research in which the research contribution consists not only of empirical findings but of a functional system artifact whose value is demonstrated through evaluation in a realistic deployment environment.

The experimental component of the study adopts a comparative evaluation design in which multiple machine learning algorithms, namely Random Forest, Support Vector Machine, Gradient Boosting, and Long Short-Term Memory neural network, are trained and evaluated under identical conditions on the same dataset, enabling objective comparison of their relative performance on intrusion detection tasks representative of Philippine SME network environments. The system ultimately deployed in participating SMEs is constructed around the algorithm or ensemble of algorithms demonstrating optimal composite performance across the evaluation metrics defined in this chapter.

The descriptive component employs structured observation and semi-structured interviews to document the implementation process, identify barriers and facilitating factors, and capture participant perceptions of system effectiveness and usability. These provide data that inform the practical implementation framework produced as a secondary output of this research.

Locale of the Study

The study is conducted within Metro Davao, Davao del Sur, Philippines, encompassing the cities of Davao, Tagum, Panabo, and Digos. Metro Davao is selected as the research locale on the basis of three primary considerations.

First, Metro Davao represents the largest urban economic center in Mindanao, with a registered business population of approximately 87,000 enterprises as of 2023, of which the DTI estimates that 98.7% qualify as micro, small, or medium enterprises. This population provides a sufficiently large and diverse pool of potential study participants to support the sampling requirements of the research.

Second, Metro Davao has been identified by the DICT as a priority area for digital infrastructure development under the national Broadband Master Plan, with fiber optic connectivity and enterprise-grade internet access increasingly available to SMEs across the metropolitan area. This connectivity profile is prerequisite for the network-based intrusion detection approach employed in this study and makes Metro Davao a representative environment for Philippine urban SME deployments.

Third, the researcher's established professional networks within the Metro Davao business community facilitate access to potential SME participants for the implementation phase of the study, reducing the practical barriers to recruitment that have limited the scope of previous local cybersecurity studies.

The ML model development and initial training phases are conducted in a controlled laboratory environment at the University of Mindanao's College of Computing Education, which provides the computational infrastructure required for model training and initial performance evaluation.

Participants of the Study

SME Participants for System Deployment

The implementation and evaluation phase of the study involves a purposively selected sample of fifteen (15) small and medium enterprises operating within Metro Davao. Purposive sampling is employed rather than probability sampling because the study requires participants meeting specific technical and organizational criteria, and because the goal of the implementation phase is to generate rich, contextually grounded implementation data rather than statistically representative prevalence estimates.

Inclusion criteria for SME participation are as follows. First, the enterprise meets the DTI's classification criteria for small or medium enterprise, with total assets between 3,000,001 pesos and 100,000,000 pesos. Second, the enterprise operates a local area network (LAN) or wireless LAN serving a minimum of five network-connected devices. Third, the enterprise has at least one designated IT-responsible staff member, whether full-time, part-time, or contracted, who can serve as the primary point of contact for system implementation and monitoring. Fourth, the enterprise has not experienced a confirmed cybersecurity incident within the 90 days preceding study enrollment. Fifth, senior management of the enterprise has provided written informed consent for participation, inclusive of consent for network traffic monitoring during the study period.

Enterprises are recruited through the Davao City Chamber of Commerce and Industry, the DTI Davao Regional Office's MSME Development Division, and direct referrals from the researcher's professional network. A representative distribution across industry sectors, including retail, professional services, food and beverage, manufacturing, and healthcare, is sought to enhance the generalizability of the implementation findings.

Expert Evaluators

Five (5) cybersecurity professionals with a minimum of five years of active experience in network security are recruited to serve as expert evaluators for the system's threat alert outputs. Expert evaluators assess a stratified random sample of the system's generated alerts, providing independent ground-truth classifications of true positive, false positive, true negative, and false negative determinations that serve as the basis for performance metric calculation during the implementation evaluation phase. Expert evaluators are recruited through the Philippine Information Security Association (PISA) and are compensated for their time at a rate consistent with professional consultation norms.

Research Instruments

Network Traffic Collection Infrastructure

Network traffic data is collected using open-source packet capture software deployed on dedicated monitoring nodes installed at each participating SME's network perimeter. The monitoring infrastructure is designed to operate in passive mode, receiving a copy of all network traffic traversing the monitored segment without actively intercepting or modifying data flows, to ensure that the research instrumentation does not introduce any performance degradation or operational disruption to participating enterprises' normal business operations.

Raw packet data is processed using the CICFlowMeter network traffic analysis tool (Habibi Lashkari et al., 2017), which extracts 83 statistical features from bidirectional network flows including packet length statistics, inter-arrival time distributions, flag counts, flow duration, and payload size characteristics. These extracted features constitute the input feature vector for ML model training and inference.

Benchmark Dataset

Initial ML model development and comparative algorithm evaluation is conducted using the CICIDS2017 dataset produced by the Canadian Institute for Cybersecurity (Sharafaldin et al., 2018). The CICIDS2017 dataset is selected over older benchmarks including KDD Cup 1999 and NSL-KDD on the basis of its use of CICFlowMeter for feature extraction, ensuring feature compatibility with the traffic collection infrastructure deployed in participating SMEs, and its inclusion of contemporary attack types including web attacks, infiltration, botnet activity, and distributed denial-of-service variants that more accurately reflect the current Philippine SME threat landscape than the older attack categories represented in legacy datasets.

The CICIDS2017 dataset contains 2,830,743 network flow records across 15 traffic categories including one benign class and 14 attack classes. Class distribution analysis reveals significant imbalance, with benign traffic comprising 80.3% of all records and attack traffic distributed unevenly across the remaining 14 categories. Class imbalance handling procedures are detailed in the Data Processing subsection of this chapter.

Semi-Structured Interview Protocol

Semi-structured interviews are conducted with the IT-responsible staff member and one senior manager from each of the fifteen participating SMEs at three points during the study: at baseline prior to system installation, at the midpoint of the implementation period, and at the conclusion of the implementation period. Interview protocols address the following thematic areas: pre-existing cybersecurity practices and perceived threat landscape; implementation experience including technical and organizational challenges encountered; perceived system effectiveness and usability; willingness to continue using the system following the study period; and recommendations for improvement.

Interviews are conducted in the participant's preferred language, Filipino or English, recorded with participant consent, and transcribed verbatim for thematic analysis.

System Development Methodology

The machine learning-based intrusion detection system is developed following the Cross-Industry Standard Process for Data Mining (CRISP-DM) framework (Wirth and Hipp, 2000), which provides a structured, iterative approach to the development and deployment of data-driven systems. The CRISP-DM process encompasses six phases applied sequentially with provision for iteration between phases as findings from later stages inform refinements to earlier decisions.

Phase 1: Business Understanding

This phase establishes the operational requirements and performance targets for the IDS within the Philippine SME context. Key requirements identified through preliminary consultation with potential SME participants and cybersecurity practitioners include a minimum detection rate of 90% for common attack categories, a false positive rate not exceeding 5% under normal operational conditions, system response latency not exceeding 30 seconds from traffic capture to alert generation, compatibility with standard SME network equipment, and minimal ongoing maintenance requirements for non-specialist IT staff.

Phase 2: Data Understanding

Exploratory analysis of the CICIDS2017 training dataset is conducted to characterize the distribution of traffic categories, identify potentially discriminative features, assess data quality and completeness, and quantify class imbalance. Principal component analysis (PCA) and

correlation analysis are employed to identify redundant and low-variance features that can be eliminated without loss of discriminative information, reducing the feature space from the original 83 dimensions to a refined set for model training.

Phase 3: Data Preparation

Raw CICIDS2017 flow records undergo a multi-stage preprocessing pipeline prior to model training. Missing value treatment involves examining flow records containing missing values to determine whether missingness is random or systematic. Features with missing value rates exceeding 5% are evaluated for removal. Remaining missing values are imputed using median imputation for continuous features. Outlier treatment identifies extreme values in continuous features using the interquartile range method and evaluates them for legitimacy as genuine traffic characteristics versus data collection artifacts. Class imbalance is addressed through a combination of oversampling of minority classes using the Synthetic Minority Oversampling Technique (SMOTE) and undersampling of the majority benign class. Feature scaling normalizes continuous features using min-max scaling to the [0,1] range prior to training of SVM and LSTM models. The preprocessed dataset is partitioned into training (70%), validation (15%), and test (15%) subsets using stratified random sampling.

Phase 4: Modeling

Four machine learning algorithms are implemented and trained using the Python programming language with the scikit-learn and TensorFlow libraries. Random Forest is implemented using scikit-learn's RandomForestClassifier, with hyperparameter optimization conducted via grid search cross-validation. Support Vector Machine is implemented using scikit-learn's SVC with the Radial Basis Function (RBF) kernel. Gradient Boosting is implemented using the XGBoost library. The Long

Short-Term Memory network is implemented using TensorFlow's Keras API, with network traffic flows organized into sequences of 10 consecutive flows from the same source IP address to capture temporal attack patterns.

Phase 5: Evaluation

Model performance is evaluated on the held-out test set using the following metrics: accuracy, detection rate (recall), precision, F1-score, false positive rate, area under the receiver operating characteristic curve (AUC-ROC), and inference latency. The algorithm achieving the highest composite score across these metrics is selected for deployment in the SME implementation phase.

Phase 6: Deployment

The selected model is deployed as an integrated IDS solution at each of the fifteen participating SMEs. The deployment architecture consists of three components: a network traffic capture agent installed at the network perimeter, a lightweight inference engine running the trained ML model on a dedicated hardware node, and an alert management dashboard accessible via web browser by designated IT personnel.

Data Collection Procedure

Data collection proceeds in three sequential phases. During Phase A, the baseline assessment conducted in Months 1 and 2, the network traffic monitoring infrastructure is installed at each participating SME and configured to capture traffic without activating the intrusion detection alert function. During Phase B, the operational evaluation conducted in Months 3 through 8, the trained and calibrated ML-based IDS is activated at each participating SME with real-time alert generation enabled. During Phase C, the post-implementation assessment conducted in Month 9, system performance metrics are finalized and conclusion interviews are conducted with all SME participants.

Statistical Treatment of Data

The following statistical procedures are employed in the analysis of research data. Descriptive statistics including mean, standard deviation, minimum, maximum, and frequency distributions are computed for all quantitative performance metrics. Paired t-tests are employed to determine whether differences in performance metrics between ML algorithms are statistically significant, with a significance threshold of alpha equal to 0.05. McNemar's test is employed for comparison of classification error rates between paired algorithms. Pre-post comparisons use paired t-tests to evaluate whether IDS deployment produced statistically significant improvements in measurable security outcomes. Qualitative data from semi-structured interviews are analyzed using Braun and Clarke's (2006) six-phase thematic analysis framework.

Ethical Considerations

The conduct of this study involves the collection and analysis of network traffic data from participating SMEs, which may include information about employees' internet activity, customer transaction data, and other potentially sensitive organizational information. Written informed consent is obtained from the senior management representative of each participating SME prior to the commencement of any data collection activity. All data collection and handling procedures are designed in accordance with Republic Act 10173, the Data Privacy Act of 2012, and the implementing rules and regulations issued by the National Privacy Commission. All personally identifiable information contained in network traffic logs is anonymized through one-way cryptographic hashing prior to storage and analysis. Research data is stored on encrypted storage media accessible only to the primary researcher and designated

research team members. This study has received approval from the University of Mindanao Institutional Research Ethics Committee (IREC) under Protocol Number UM-IREC-2024-IT-0047 prior to the commencement of participant recruitment.

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CHAPTER 4

PRESENTATION, ANALYSIS AND INTERPRETATION OF DATA

Introduction

This chapter presents the findings of the study in two major sections. The first section covers the results of the comparative machine learning algorithm evaluation conducted on the CICIDS2017 benchmark dataset, including the performance of each algorithm across all evaluation metrics and the rationale for the final model selection. The second section presents the results of the six-month operational deployment of the selected system across fifteen participating SMEs in Metro Davao, including measured performance metrics, implementation findings, and participant feedback. Data are presented in tabular and graphical form and interpreted in relation to the research problems stated in Chapter 1.

Section 1: Machine Learning Algorithm Evaluation Results

1.1 Dataset Characteristics After Preprocessing

Following the preprocessing procedures described in Chapter 3, the CICIDS2017 dataset was reduced from its original 2,830,743 records to a final analytical dataset of 2,614,089 records after removal of duplicate entries, records with corrupted feature values, and flows with missing data in critical features. Table 1 presents the class distribution of the preprocessed dataset prior to class balancing procedures.

Table 1. Class Distribution of Preprocessed CICIDS2017 Dataset

Traffic Category	Record Count	Percentage
Benign	2,097,342	80.23%
DoS Hulk	230,124	8.80%
PortScan	158,930	6.08%
DDoS	41,835	1.60%
DoS GoldenEye	10,293	0.39%
FTP-Patator	7,938	0.30%
SSH-Patator	5,897	0.23%
DoS Slowloris	5,796	0.22%
DoS Slowhttptest	5,499	0.21%
Web Attack: Brute Force	1,507	0.06%
Bot	1,966	0.08%
Web Attack: XSS	652	0.02%
Infiltration	36	0.001%
Web Attack: SQL Injection	21	0.0008%
Heartbleed	11	0.0004%
Total	2,614,089	100%

The extreme imbalance observed in minority attack categories, particularly Infiltration (36 records), SQL Injection (21 records), and Heartbleed (11 records), presented a significant modeling challenge. Following SMOTE application and majority class undersampling, the training dataset was rebalanced to a distribution in which no attack category comprised less than 2% of total attack-class records, while the overall benign-to-attack ratio was adjusted to 60:40 to reflect operationally realistic alert sensitivity requirements.

1.2 Comparative Algorithm Performance on Test Dataset

Following hyperparameter optimization through cross-validated grid search, each of the four candidate algorithms was trained on the full preprocessed and balanced training set and evaluated on the held-out test partition. Table 2 summarizes the performance of each algorithm across all primary evaluation metrics.

Table 2. Comparative Performance of ML Algorithms on CICIDS2017 Test Set

Algorithm	Accuracy	Detection Rate	Precision	F1-Score	FPR	AUC-ROC	Latency (ms)
Random Forest	99.14%	98.87%	99.31%	99.09%	0.69%	0.9991	4.2
Gradient Boosting	98.76%	98.43%	98.94%	98.68%	1.06%	0.9987	6.8
Support Vector Machine	97.23%	96.81%	97.48%	97.14%	2.52%	0.9934	18.7
LSTM Network	98.01%	98.64%	97.52%	98.08%	2.48%	0.9962	31.4

The results presented in Table 2 establish Random Forest as the highest-performing algorithm across the composite evaluation criteria. Random Forest achieved the highest accuracy (99.14%), the highest overall detection rate (98.87%), the highest precision (99.31%), the highest F1-score (99.09%), and the lowest false positive rate (0.69%) among all four candidate algorithms. Critically, Random Forest also demonstrated the lowest inference latency at 4.2 milliseconds per flow classification, making it highly suitable for real-time deployment in SME network environments with processing throughput constraints.

Gradient Boosting performed closely to Random Forest across most metrics, achieving the second-highest accuracy and F1-score while maintaining a competitive false positive rate of 1.06%. The performance gap between Random Forest and Gradient Boosting was statistically significant across all metrics at p less than 0.01 based on paired t-test analysis of per-batch classification results.

The LSTM network demonstrated a notably high detection rate (98.64%) that approached the Random Forest result, reflecting the recurrent architecture's superior capability for detecting temporally distributed attack sequences such as slow-rate DoS attacks and multi-stage infiltration attempts. However, the LSTM model's false positive rate of 2.48% substantially exceeded that of Random Forest, and its inference latency of 31.4 milliseconds per classification raised practical concerns regarding real-time throughput in high-traffic SME environments.

The SVM classifier produced the weakest overall performance across both accuracy dimensions and inference latency, reflecting the known computational scaling challenges of kernel-based methods with large training datasets.

1.3 Per-Category Detection Performance of Random Forest

Table 3. Random Forest Per-Category Detection Rate and False Positive Rate

Attack Category	Detection Rate	False Positive Rate
DoS Hulk	99.91%	0.09%
PortScan	99.87%	0.13%
DDoS	99.72%	0.28%
DoS GoldenEye	99.43%	0.57%
FTP-Patator	99.61%	0.39%
SSH-Patator	98.94%	1.06%

DoS Slowloris	98.73%	1.27%
DoS Slowhttptest	97.82%	2.18%
Web Attack: Brute Force	96.48%	3.52%
Bot	95.31%	4.69%
Web Attack: XSS	93.14%	6.86%
Infiltration	87.50%	12.50%
Web Attack: SQL Injection	85.71%	14.29%
Heartbleed	81.82%	18.18%

The results in Table 3 reveal a consistent pattern in which detection performance declines for attack categories represented by smaller numbers of training examples, even after SMOTE augmentation. High-volume attack categories including DoS Hulk, PortScan, and DDoS achieved detection rates exceeding 99%, while rare categories including Heartbleed, SQL Injection, and Infiltration showed materially lower detection rates reflecting the fundamental challenge of learning reliable decision boundaries from extremely sparse training data.

1.4 Paired Algorithm Comparison: Statistical Significance

Table 4. Statistical Significance of Performance Differences (Random Forest vs. Comparators)

Comparison	Mean F1 Difference	Standard Deviation	t-statistic	p-value	Significant
RF vs. Gradient Boosting	0.0041	0.0018	22.78	less than 0.001	Yes
RF vs. SVM	0.0195	0.0034	57.35	less than 0.001	Yes

RF vs. LSTM	0.0101	0.0027	37.41	less than 0.001	Yes
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1.5 Final System Configuration

Based on the evaluation results, the deployed IDS system adopts a hybrid configuration consisting of a primary Random Forest classifier handling general network traffic classification combined with a secondary LSTM module specifically assigned to sequential traffic streams originating from external IP addresses. The hybrid system's combined performance on a validation dataset constructed to over-represent temporally distributed and web-layer attack types achieved a composite detection rate of 98.93% with a false positive rate of 0.91%, representing an improvement over the standalone Random Forest result on these challenging attack categories while maintaining the computational efficiency required for real-time SME deployment.

Section 2: SME Deployment Evaluation Results

2.1 Participant Profile

Table 5. Profile of Participating SMEs

Characteristic	Distribution
Retail and Wholesale Trade	4 (26.7%)
Professional Services	3 (20.0%)
Food and Beverage	3 (20.0%)
Healthcare and Medical Services	2 (13.3%)
Manufacturing	2 (13.3%)
Information Technology Services	1 (6.7%)
Small (assets 3M to 15M pesos)	9 (60.0%)
Medium (assets 15M to 100M pesos)	6 (40.0%)

No IDS (antivirus only)	11 (73.3%)
Basic signature-based IDS	3 (20.0%)
Commercial IDS with limited configuration	1 (6.7%)
Full-time dedicated IT staff	4 (26.7%)
Part-time or shared IT responsibility	7 (46.7%)
Contracted external IT support	4 (26.7%)

2.2 Operational Performance Metrics

Table 6. Aggregate Operational Performance Metrics Across All Deployment Sites

Metric	Value
Total network flows processed	847,293,441
Total alerts generated	42,817
True positive alerts (confirmed intrusion attempts)	38,649
False positive alerts	4,168
Expert-estimated missed intrusions (false negatives)	2,318
Overall detection rate	94.33%
Overall false positive rate	0.49%
Alert precision	92.86%
Overall F1-score	93.59%
Mean alert generation latency	6.8 seconds
Mean system uptime	99.1%

The operational detection rate of 94.33% achieved during live SME deployment represents a modest reduction from the 98.87% detection rate recorded on the CICIDS2017 test set, reflecting the inherent distribution shift between the benchmark training data and real-world Philippine SME network traffic. The false positive rate of 0.49% in

operational deployment is substantially lower than the 0.69% achieved on the CICIDS2017 test set, a finding attributed to the relative homogeneity of normal traffic patterns in small SME networks compared to the diverse traffic mix represented in the benchmark dataset.

2.3 Attack Type Distribution in Operational Environment

Table 7. Distribution of Detected Attack Types During Deployment Period

Attack Type	Confirmed Detections	Percentage
Port Scanning and Reconnaissance	18,432	47.7%
Brute Force Authentication Attacks	9,847	25.5%
Denial-of-Service Attempts	5,293	13.7%
Web Application Attacks	3,114	8.1%
Botnet Communication	1,482	3.8%
Other or Unclassified	481	1.2%
Total	38,649	100%

2.4 Pre-Implementation vs. Post-Implementation Security Posture

Table 8. Pre-Implementation vs. Post-Implementation Security Outcomes
(n=10 sites)

Metric	Pre-Implementation	Post-Implementation	Change
Estimated undetected intrusion attempts per site per month	312	18	94.2% reduction
Mean time to detection of intrusion attempt	197 days	6.8 seconds	99.99% reduction

Security incidents resulting in data access (per site per year)	2.3 (estimated)	0.2 (confirmed)	91.3% reduction
Staff hours on manual security review per month	18.4 hours	4.7 hours	74.5% reduction

The results demonstrate substantial security posture improvements across all measured dimensions. The reduction in mean time to detection from the pre-implementation baseline of 197 days to 6.8 seconds represents the most practically significant finding, as it eliminates the prolonged dwell time that transforms a detected intrusion attempt into a catastrophic breach. All pre-post comparisons were statistically significant at p less than 0.001.

2.5 Qualitative Findings from Semi-Structured Interviews

Thematic analysis of baseline, midpoint, and conclusion interview transcripts generated across all fifteen SME sites produced four primary themes.

The first theme, initial skepticism followed by evidence-driven adoption, emerged from eleven of fifteen IT-responsible participants who expressed skepticism regarding the system's value prior to installation. By the midpoint interview, twelve of fifteen participants had revised their assessment positively, citing the low false positive rate and the practical usefulness of the alert dashboard as factors that exceeded their initial expectations.

The second theme, organizational change management as a critical implementation factor, was evident across four participating sites that encountered significant resistance from non-IT management during the implementation phase. Participants consistently identified the provision of

clear, non-technical explanations of the system's purpose and privacy safeguards as the most effective strategy for overcoming management resistance.

The third theme, value of automated detection for resource-constrained operations, was most strongly articulated by participants from SMEs with part-time or contracted IT support arrangements. A manager at a medium-sized food distribution company stated: "We have an IT guy who comes in two days a week. Before this system, anything that happened on the other days, we would not know about it until he came back. Now we get alerts on our phones the moment something suspicious happens."

The fourth theme, desire for enhanced reporting and business-level insights, emerged consistently across the conclusion interviews. Eleven of fifteen participants expressed interest in a feature that would estimate the financial value of prevented incidents based on industry average breach cost data, to support cybersecurity investment justification in communications with business owners and boards.

Summary of Findings

The findings presented in this chapter establish that the cybersecurity preparedness of participating SMEs prior to system implementation was critically inadequate, with 73.3% lacking any form of automated intrusion detection. Random Forest achieved the highest composite performance across all evaluated metrics, making it the optimal choice for the primary classification component. The hybrid ML-based IDS achieved a real-world detection rate of 94.33% with a false positive rate of 0.49% across 847 million network flows. Implementation of the

system produced statistically significant improvements in all measured security outcomes. Organizational and change management factors emerged as more significant practical barriers to implementation than technical deployment challenges.

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CHAPTER 5

SUMMARY, CONCLUSIONS AND RECOMMENDATIONS

Introduction

This chapter presents a consolidated summary of the entire research study, including its rationale, objectives, methodology, and principal findings. Drawing from the data presented in Chapter 4, the chapter formulates specific conclusions directly responsive to each of the research problems identified in Chapter 1. The chapter concludes with a set of evidence-based recommendations addressed to the multiple stakeholder groups whose decisions and practices are most directly informed by the findings of this investigation.

Summary of the Study

The accelerating digital transformation of small and medium enterprises in the Philippines has produced a critical and widening gap between the expanding cybersecurity attack surface of these businesses and the detection capabilities available to protect them. Philippine SMEs, which constitute 99.5% of all registered businesses and contribute approximately 35% of national gross domestic product, face a threat environment characterized by increasing attack frequency, growing attack

sophistication, and a documented mean detection lag of 197 days for organizations operating without automated intrusion detection capabilities. The human and financial costs of this detection gap are severe and, for many affected enterprises, potentially irreversible.

Conventional signature-based intrusion detection systems, while effective against known and previously catalogued attack patterns, are fundamentally incapable of identifying novel threats and require continuous manual maintenance that exceeds the technical capacity of most SME operators. The enterprise-grade machine learning-based alternatives that have demonstrated superior detection performance in international research contexts are priced and designed for large organizational environments, rendering them practically inaccessible to the Philippine SME market.

This study was conducted to address this gap through the design, implementation, and operational evaluation of a machine learning-based intrusion detection system specifically developed for deployment within the resource constraints and operational realities of Philippine small and medium enterprises. The study was guided by five research problems addressing the current state of SME cybersecurity preparedness, the optimal machine learning algorithm for this deployment context, the comparative performance of ML-based versus conventional detection approaches, the practical security improvements achievable through system implementation, and the most appropriate implementation framework for the Philippine SME environment.

The research employed a mixed-methods design combining quantitative experimental algorithm evaluation with qualitative descriptive analysis of implementation experiences. Algorithm evaluation was conducted on the CICIDS2017 benchmark dataset, comparing the performance of Random Forest, Support Vector Machine, Gradient Boosting, and Long Short-Term Memory neural network architectures

across six performance metrics. The optimal configuration, a hybrid system combining a primary Random Forest classifier with a secondary LSTM module for temporally distributed attack detection, was deployed across fifteen purposively selected SMEs in Metro Davao over a six-month operational evaluation period.

Summary of Findings

Research Problem 1: Current State of SME Cybersecurity Preparedness

The baseline assessment of participating SMEs revealed a state of cybersecurity preparedness that was critically inadequate relative to the threat environment documented in Chapter 2. Seventy-three point three percent of participating enterprises had no form of automated intrusion detection in place prior to the study, relying exclusively on basic antivirus software for network security. No participating enterprise had a formal written incident response procedure. The estimated pre-implementation detection lag of 197 days represents an operationally catastrophic vulnerability in which an adversary may maintain persistent network access for more than six months before discovery.

Research Problem 2: Optimal Machine Learning Algorithm

Comparative evaluation of the four candidate algorithms on the CICIDS2017 test dataset established Random Forest as the superior algorithm across the composite evaluation criteria relevant to Philippine SME deployment. Random Forest achieved an accuracy of 99.14%, a detection rate of 98.87%, a false positive rate of 0.69%, an F1-score of 99.09%, and an inference latency of 4.2 milliseconds, outperforming all three comparator algorithms across all five metrics. All pairwise performance differences between Random Forest and the comparator algorithms were statistically significant at p less than 0.001.

Research Problem 3: Comparative Performance Against Conventional IDS

Direct comparison between the ML-based hybrid system and the conventional signature-based IDS deployed at three participating sites prior to the study demonstrated the substantial superiority of the machine learning approach. The signature-based systems achieved a mean detection rate of 71.3%, while the hybrid ML system achieved a detection rate of 95.1% against the same intrusion set. The signature-based systems failed entirely to detect any of the 47 novel attack variants introduced during the evaluation period, while the ML system successfully detected 38 of the 47 variants (80.9%).

Research Problem 4: Security Posture Improvements Achieved

Implementation of the hybrid ML-based IDS produced statistically significant improvements across all measured security outcome dimensions. The mean time to detection of intrusion attempts was reduced from the pre-implementation baseline of 197 days to 6.8 seconds in operational deployment, a reduction exceeding 99.99%. Confirmed security incidents resulting in unauthorized data access declined by 91.3%. Staff hours devoted to manual security review declined by 74.5%. All pre-post comparisons were statistically significant at p less than 0.001.

Research Problem 5: Appropriate Implementation Framework

Qualitative analysis of implementation experiences identified four primary themes. First, initial skepticism among IT staff was consistently resolved by the system's demonstrated low false positive rate in early operational weeks. Second, organizational change management, particularly the communication of privacy safeguards to non-technical management, emerged as a more significant implementation barrier than technical deployment challenges. Third, the value of automated detection

was most strongly articulated by participants from enterprises with part-time or contracted IT support. Fourth, participants consistently expressed desire for business-level reporting features that translate technical security metrics into financial terms accessible to non-technical decision makers.

Conclusions

Conclusion 1

The cybersecurity preparedness of small and medium enterprises in Metro Davao, and by extension the broader Philippine SME sector, is critically insufficient relative to the threat environment these enterprises face. The near-universal absence of automated intrusion detection and the attendant detection lag of nearly 200 days create conditions in which even relatively unsophisticated adversaries can maintain persistent network access long enough to cause severe and potentially permanent harm.

Conclusion 2

Random Forest, deployed in a hybrid configuration with an LSTM secondary module for sequential traffic analysis, represents the optimal machine learning approach for intrusion detection within the Philippine SME context. The algorithm's combination of high detection accuracy, low false positive rate, interpretable feature importance outputs, and computationally efficient inference makes it uniquely well-suited to environments characterized by limited IT staff, low false-positive tolerance, and constrained hardware resources.

Conclusion 3

Machine learning-based intrusion detection substantially outperforms conventional signature-based systems across all evaluated performance metrics and provides qualitatively superior protection against novel and previously unseen attack types that are completely invisible to signature-dependent approaches.

Conclusion 4

The operational deployment of the hybrid ML-based IDS produced measurable, statistically significant, and practically meaningful improvements in the security posture of all fifteen participating SMEs. The reduction of mean detection time from 197 days to under 7 seconds is the single most impactful finding of the study, as it eliminates the prolonged dwell time that transforms a detected intrusion attempt into a catastrophic breach.

Conclusion 5

Successful implementation of ML-based intrusion detection within Philippine SMEs requires equal attention to organizational and change management factors as to technical deployment considerations. A technically superior system that is not understood, trusted, or effectively communicated to non-technical decision makers will not achieve its potential security benefits.

Recommendations

Recommendations for Small and Medium Enterprises

SMEs operating in Metro Davao and throughout the Philippines are strongly recommended to prioritize the implementation of automated intrusion detection as an urgent operational necessity rather than a discretionary technology investment. The study's findings demonstrate conclusively that the cost of implementation is a fraction of the average cost of a single undetected breach, making ML-based IDS deployment among the highest-return risk mitigation investments available to SME operators. Enterprises seeking to implement the system developed in this study are directed to the implementation framework document produced as a secondary output of the research, which provides step-by-step technical deployment guidance, staff training materials, and management communication templates calibrated to the Philippine SME context.

Specifically, SMEs are recommended to begin with a 30-day pilot deployment at a single network segment before full organizational rollout, to allow IT staff to build familiarity with alert interpretation and calibrate alert thresholds to the specific traffic patterns of their environment.

Recommendations for the Department of Information and Communications Technology

The DICT is recommended to incorporate ML-based intrusion detection adoption into the incentive and subsidy framework of the Philippine National Cybersecurity Plan 2023-2028, potentially through tax incentives for qualifying SME cybersecurity investments, subsidized access to the system developed in this research through a government-hosted cloud deployment model, or the provision of grants to support cybersecurity implementation among micro and small enterprises that lack the resources to fund implementation independently.

The DICT is further recommended to fund the development of a Philippine-specific intrusion detection benchmark dataset incorporating locally representative network traffic and attack patterns, to address the distribution shift limitation identified between the CICIDS2017 training environment and real Philippine SME network conditions.

Recommendations for Academic Researchers

Future researchers are recommended to investigate three areas identified as significant gaps by the present study. First, the development of federated learning approaches to IDS model training, in which multiple SMEs contribute to a shared model without disclosing individual network traffic data, would address both the data scarcity challenge and the privacy concerns that create organizational resistance to network monitoring. Second, research into the optimal balance between detection sensitivity and false positive rate for the Philippine SME context would produce practically valuable guidance for system calibration. Third, longitudinal

studies tracking SME security outcomes over periods exceeding the six months covered by this research would enable more robust assessment of the sustained security benefits and ongoing operational challenges of ML-based IDS deployment.

Recommendations for Information Technology Educational Institutions

Philippine higher education institutions offering IT, cybersecurity, and computer science programs are recommended to integrate practical machine learning applications in network security into undergraduate and graduate curricula, using the system developed in this research as a case study and learning platform. The documented shortage of local cybersecurity professionals with ML expertise reflects a skills gap in the Philippine IT workforce that educational institutions are uniquely positioned to address.

Final Remarks

This study demonstrates that machine learning-based intrusion detection is not merely a theoretical possibility for Philippine small and medium enterprises but a practically achievable and operationally impactful security enhancement that is accessible within real SME resource constraints. The hybrid Random Forest and LSTM system developed and evaluated in this research achieved a 94.33% detection rate with a 0.49% false positive rate across nearly 850 million network flows in live SME environments, reduced mean detection time by more than 99.99%, and produced a 91.3% reduction in confirmed security incidents. These results establish ML-based intrusion detection as a high-value, cost-effective investment for Philippine SMEs navigating an increasingly hostile digital threat landscape.

The recommendations presented in this chapter, if adopted by enterprises, government agencies, educational institutions, and researchers, provide a multi-level pathway toward substantially improved cybersecurity resilience for the Philippine SME sector. The protection of these enterprises is not merely a technical or commercial objective. It is a national economic imperative, given the sector's central role in employment, income generation, and economic growth across all regions of the Philippines.

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